

## ISACHSEN, NU, Canada

WMO: 710740

Lat: 78.792N

Lon: 103.554W

Elev: 190

StdP: 14.60

Time Zone: -7.00 (NAM)

Period: 96-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |      |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|------|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%   |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS   | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l)  | (m)   | (n)                   | (o)  | (p)   |
| 2             | -46.9      | -44.1 | -54.7                         | 0.2 | -46.6 | -52.1 | 0.3 | -44.0 | 48.3                  | -15.2 | 43.4 | -16.9 | 4.2                   | 220  | 1.226 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 7.8                    | 53.6            | 47.2 | 50.0 | 45.0 | 46.5 | 42.9 | 47.5                | 53.1 | 45.4 | 49.9 | 43.2 | 46.7 | 10.8                 | 20   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 43.3                            | 41.7 | 49.4 | 41.4 | 38.7 | 47.5 | 39.5 | 35.9 | 45.4 | 18.9          | 53.5 | 17.9 | 50.0 | 16.8 | 46.9 | 54.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 40.4          | 34.3 | 28.5 | DB                         | -50.5 | 54.6               | 4.6 | 7.3                                                | -53.9 | 59.8       | -56.6 | 64.0       | -59.2 | 68.1       | -62.5 | 73.4 | (4) |
| (5)               |      |      | WB                         | -49.7 | 48.2               | 0.8 | 3.9                                                | -50.2 | 51.0       | -50.7 | 53.3       | -51.1 | 55.5       | -51.7 | 58.3 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 1.8    | -26.8 | -27.6 | -25.2 | -8.8  | 13.7  | 32.1  | 38.9  | 33.3  | 19.4  | 1.6   | -10.4 | -21.0 |  |  |
|                                                                        | DBStd       | 25.54  | 8.34  | 9.44  | 10.42 | 8.76  | 8.89  | 6.64  | 6.24  | 5.98  | 7.77  | 10.23 | 9.95  | 9.98  |  |  |
|                                                                        | HDD50       | 17615  | 2381  | 2171  | 2331  | 1764  | 1125  | 539   | 350   | 520   | 919   | 1501  | 1813  | 2201  |  |  |
|                                                                        | HDD65       | 23083  | 2846  | 2591  | 2796  | 2214  | 1590  | 988   | 810   | 984   | 1369  | 1966  | 2263  | 2666  |  |  |
|                                                                        | CDD50       | 6      | 0     | 0     | 0     | 0     | 0     | 1     | 5     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 11.5   | 11.9  | 10.4  | 10.3  | 10.0  | 12.1  | 12.3  | 12.2  | 11.9  | 11.8  | 11.1  | 11.4  | 12.4  |  |  |
|                                                                        | PrecAvg     | 4.9    | 0.2   | 0.2   | 0.2   | 0.2   | 0.5   | 0.5   | 0.9   | 0.9   | 0.7   | 0.5   | 0.2   | 0.2   |  |  |
| (15) Precipitation                                                     | PrecMax     | 8.1    | 0.6   | 0.6   | 0.8   | 0.6   | 1.1   | 1.3   | 2.1   | 2.0   | 2.4   | 1.0   | 0.4   | 0.5   |  |  |
|                                                                        | PrecMin     | 2.7    | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.1   | 0.2   | 0.3   | 0.0   | 0.0   | 0.0   | 0.0   |  |  |
|                                                                        | PrecStd     | 1.5    | 0.2   | 0.2   | 0.2   | 0.2   | 0.2   | 0.4   | 0.6   | 0.5   | 0.6   | 0.3   | 0.1   | 0.2   |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | -3.2  | -0.3  | 7.0   | 14.1  | 34.7  | 55.2  | 59.5  | 53.6  | 36.3  | 26.9  | 18.8  | 9.5   |  |  |
|                                                                        |             | MCWB   | -3.7  | -0.9  | 6.4   | 13.1  | 30.7  | 48.5  | 49.2  | 48.2  | 34.7  | 26.9  | 18.5  | 9.0   |  |  |
|                                                                        | 2%          | DB     | -7.1  | -6.3  | -1.0  | 9.6   | 30.5  | 49.0  | 55.7  | 48.7  | 33.5  | 23.8  | 12.4  | 3.4   |  |  |
|                                                                        |             | MCWB   | -7.5  | -6.6  | -1.4  | 9.0   | 29.1  | 43.9  | 48.0  | 44.6  | 32.4  | 23.5  | 11.7  | 3.0   |  |  |
|                                                                        | 5%          | DB     | -10.9 | -9.3  | -6.5  | 6.5   | 27.9  | 44.6  | 52.4  | 45.0  | 31.4  | 20.6  | 8.0   | -1.3  |  |  |
|                                                                        |             | MCWB   | -11.2 | -9.6  | -7.0  | 5.8   | 27.0  | 40.4  | 46.3  | 42.7  | 30.8  | 20.3  | 7.5   | -1.7  |  |  |
|                                                                        | 10%         | DB     | -14.6 | -12.7 | -11.0 | 2.8   | 25.5  | 41.2  | 49.5  | 41.8  | 29.5  | 16.8  | 4.0   | -7.2  |  |  |
|                                                                        |             | MCWB   | -15.0 | -13.0 | -11.3 | 2.3   | 24.8  | 38.1  | 44.6  | 40.0  | 29.1  | 16.4  | 3.5   | -7.6  |  |  |
|                                                                        | 0.4%        | WB     | -3.7  | 0.0   | 7.0   | 13.4  | 32.0  | 48.6  | 50.4  | 49.0  | 35.1  | 26.9  | 17.1  | 9.2   |  |  |
|                                                                        |             | MCDB   | -3.3  | 0.6   | 7.5   | 13.9  | 34.5  | 53.9  | 57.8  | 53.6  | 36.0  | 26.9  | 17.6  | 9.8   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | -9.0  | -6.4  | -0.9  | 9.3   | 29.3  | 44.1  | 48.3  | 45.2  | 33.0  | 23.8  | 9.4   | 1.7   |  |  |
|                                                                        |             | MCDB   | -8.7  | -6.2  | -0.6  | 9.7   | 30.6  | 48.9  | 54.9  | 48.1  | 33.7  | 23.9  | 9.9   | 2.0   |  |  |
|                                                                        | 5%          | WB     | -12.3 | -10.0 | -7.4  | 6.0   | 27.0  | 41.0  | 46.7  | 43.0  | 31.1  | 20.7  | 5.0   | -4.0  |  |  |
|                                                                        |             | MCDB   | -12.0 | -9.6  | -7.1  | 6.5   | 27.9  | 44.4  | 52.2  | 45.4  | 31.6  | 20.9  | 5.5   | -3.7  |  |  |
|                                                                        | 10%         | WB     | -16.6 | -14.5 | -12.3 | 2.3   | 24.7  | 38.3  | 44.9  | 40.4  | 29.4  | 16.9  | 0.9   | -8.6  |  |  |
|                                                                        |             | MCDB   | -16.4 | -14.2 | -12.0 | 2.7   | 25.4  | 41.2  | 49.5  | 42.2  | 29.8  | 17.1  | 1.3   | -8.3  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 9.1   | 10.1  | 10.6  | 11.0  | 8.1   | 6.1   | 7.8   | 6.5   | 8.1   | 10.3  | 10.4  | 10.3  |  |  |
|                                                                        |             | MCDBR  | 12.9  | 13.0  | 15.3  | 13.1  | 8.9   | 10.1  | 12.2  | 10.5  | 7.5   | 13.2  | 13.3  | 15.2  |  |  |
|                                                                        |             | MCWBR  | 13.6  | 14.5  | 15.9  | 13.1  | 7.6   | 6.8   | 7.7   | 7.5   | 6.8   | 13.4  | 14.6  | 17.1  |  |  |
|                                                                        |             | MCDBR  | 13.6  | 14.4  | 15.9  | 13.2  | 7.9   | 10.2  | 12.0  | 10.3  | 6.9   | 12.8  | 13.1  | 16.8  |  |  |
|                                                                        | 5% WB       | MCWBR  | 13.6  | 14.3  | 15.8  | 13.1  | 7.0   | 7.0   | 7.9   | 7.5   | 6.5   | 13.0  | 12.9  | 16.8  |  |  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | N/A   | N/A   | 0.210 | 0.276 | 0.302 | 0.329 | 0.342 | 0.297 | 0.247 | N/A   | N/A   | N/A   |  |  |
|                                                                        | taud        |        | N/A   | N/A   | 2.059 | 2.065 | 2.092 | 2.148 | 2.248 | 2.358 | 2.116 | N/A   | N/A   | N/A   |  |  |
|                                                                        | Ebn at Noon |        | 0     | 0     | 193   | 239   | 256   | 253   | 243   | 237   | 174   | 0     | 0     | 0     |  |  |
|                                                                        | Edh at Noon |        | 0     | 0     | 15    | 29    | 34    | 35    | 30    | 22    | 15    | 0     | 0     | 0     |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 0     | 0     | 225   | 1096  | 1705  | 2010  | 1696  | 924   | 314   | 0     | 0     | 0     |  |  |
|                                                                        | RadStd      |        | 0     | 0     | 25    | 74    | 133   | 198   | 150   | 91    | 27    | 0     | 0     | 0     |  |  |

## Historical Trends

|                             |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (d)                         | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)   |       |       |
| (46) Station Only           | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (0 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page